Example 6.26

Find the vector equation in parametric form and Cartesian equations of the line passing through $(-4, 2, -3)$ and is parallel to the line $\frac{-x-2}{4} = \frac{y+3}{-2} = \frac{2z-6}{3}$.

Solution

Rewriting the given equations as $\frac{x+2}{-4} = \frac{y+3}{-2} = \frac{z-3}{3/2}$ and comparing with $\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3}$, we have $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k} = -4\hat{i} - 2\hat{j} + \frac{3}{2}\hat{k} = -\frac{1}{2}(8\hat{i} + 4\hat{j} - 3\hat{k})$. Clearly, $\vec{b}$ is parallel to the vector $8\hat{i} + 4\hat{j} - 3\hat{k}$. Therefore, a vector equation of the required straight line passing through the given point $(-4, 2, -3)$ and parallel to the vector $8\hat{i} + 4\hat{j} - 3\hat{k}$ in parametric form is

$$\vec{r} = (-4\hat{i} + 2\hat{j} - 3\hat{k}) + t(8\hat{i} + 4\hat{j} - 3\hat{k}),\ t \in \mathbb{R}.$$

Therefore, Cartesian equations of the required straight line are given by

$$\frac{x+4}{8} = \frac{y-2}{4} = \frac{z+3}{-3}.$$

Example 6.27

Find the vector equation in parametric form and Cartesian equations of a straight passing through the points $(-5, 7, -4)$ and $(13, -5, 2)$. Find the point where the straight line crosses the xy-plane.

Solution

The straight line passes through the points $(-5, 7, -4)$ and $(13, -5, 2)$, and therefore, direction ratios of the straight line joining these two points are $18, -12, 6$. That is $3, -2, 1$.

So, the straight line is parallel to $3\hat{i} - 2\hat{j} + \hat{k}$. Therefore,

- Required vector equation of the straight line in parametric form is $\vec{r} = (-5\hat{i} + 7\hat{j} - 4\hat{k}) + t(3\hat{i} - 2\hat{j} + \hat{k})$ or $\vec{r} = (13\hat{i} - 5\hat{j} + 2\hat{k}) + s(3\hat{i} - 2\hat{j} + \hat{k})$ where $s, t \in \mathbb{R}$.
- Required cartesian equations of the straight line are $\frac{x+5}{3} = \frac{y-7}{-2} = \frac{z+4}{1}$ or $\frac{x-13}{3} = \frac{y+5}{-2} = \frac{z-2}{1}$.

An arbitrary point on the straight line is of the form

$$(3t - 5, -2t + 7, t - 4) \text{ or } (3s + 13, -2s - 5, s + 2)$$

Since the straight line crosses the xy-plane, the z-coordinate of the point of intersection is zero. Therefore, we have $t - 4 = 0$, that is, $t = 4$, and hence the straight line crosses the xy-plane at $(7, -1, 0)$.

Example 6.28

Find the angle between the straight line $\frac{x+3}{2} = \frac{y-1}{2} = -z$ with coordinate axes.

Solution

If $\hat{b}$ is a unit vector parallel to the given line, then $\hat{b} = \frac{2\hat{i} + 2\hat{j} - \hat{k}}{|2\hat{i} + 2\hat{j} - \hat{k}|} = \frac{1}{3}(2\hat{i} + 2\hat{j} - \hat{k})$. Therefore, from the definition of direction cosines of $\hat{b}$, we have

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